

Criterion A – Knowing and Understanding

Q1. Accuracy and Precision

Question:

Two students measure the length of the same object, whose accepted length is **20.0 cm**.

- Student A: 18.2 cm, 18.3 cm, 18.2 cm
- Student B: 19.8 cm, 20.2 cm, 20.0 cm

Outline which student's measurements are more precise and which are more accurate. Give a reason for each answer. [2]

Answer:

Student A's measurements are more **precise** because the values **18.2, 18.3 and 18.2 cm are very close to one another**.

Student B's measurements are more **accurate** because they are closer to the **accepted value of 20.0 cm**.

Remember:

Precision = closeness of repeated measurements to each other.

Accuracy = closeness of a measurement to the accepted/true value.

Q2. Complete the statements using appropriate scientific terms. [4]

a. The SI unit of length is the _____.

Answer: metre (m)

b. The amount of matter in an object is called _____.

Answer: mass

c. The SI unit of temperature is _____.

Answer: kelvin (K)

d. Measurements that are close to the true or accepted value are described as

-----.

Answer: accurate

Kelvin (K) is the SI unit of temperature

- **We commonly measure temperature in degrees Celsius (°C), for example, room temperature is about 25°C.**
- **In science, the official SI unit for temperature is kelvin (K).**
- **The Kelvin scale begins at absolute zero, the lowest possible temperature: 0 K = -273.15°C.**
- **A change of 1 K is the same size as a change of 1°C.**
- **Kelvin is written K, not °K.**

Converting Celsius to Kelvin

Temperature in K = Temperature in °C + 273.15

Q3. Rounding Measurements

Question:

A student measures the volume of water as **36.786 mL**.

Apply your scientific knowledge to **round the measurement to two decimal places** and describe why measurements should be rounded appropriately when recording experimental data. [2]

Answer:

36.786 mL = 36.79 mL

The third decimal digit is **6**, so the second decimal digit is rounded up.

Measurements should be rounded appropriately so that the recorded value reflects the **precision of the measuring instrument** and does not suggest greater precision than was actually measured.

Scientific terminology: decimal places • measurement • precision • measuring instrument

Q4. Physical and Chemical Changes

Question:

A student observes:

Situation
n

Observation

A Candle wax melts when heated and becomes solid again when cooled.

B A piece of magnesium burns and forms a white powder.

Interpret the observations to identify the physical change and chemical change. Use evidence to justify your answer. [3]

Answer:

Situation A is a physical change. Candle wax changes from a **solid to a liquid** when it melts, but **no new substance is formed**. It can become solid again when cooled.

Situation B is a chemical change. Magnesium burns and produces a **new substance**, observed as a white powder.

Therefore, **melting wax is a physical change**, while **burning magnesium is a chemical change**.

Scientific terminology: physical change • chemical change • new substance • melting • change of state

Q5. Changes of State

Question:

Three observations are recorded:

Substance	Initial state	Final state
Ice	Solid	Liquid
Water droplets on a cold glass	Gas in air	Liquid
Dry ice	Solid	Gas

A student states:

“Every substance must pass through the liquid state when changing from a solid to a gas.”

Interpret the information and make a scientifically supported judgment about the statement. Use evidence and appropriate changes-of-state terminology. [4]

Answer:

The student's statement is **incorrect**.

Ice changes from **solid to liquid**, which is called **melting**.

Water vapour changes from **gas to liquid**, which is called **condensation**.

However, dry ice changes **directly from a solid to a gas** without becoming a liquid. This change is called **sublimation**.

Therefore, the evidence shows that a substance **does not always have to pass through the liquid state** when changing from a solid to a gas.

Scientific terminology: melting • condensation • sublimation • solid • liquid • gas

Criterion B – Inquiring and Designing

Q6. Investigating the Melting of Ice

A student notices that an ice cube melts more quickly in a warm room than in a cool room. The student wants to investigate **how temperature affects the time taken for an ice cube to melt**.

Q6.1 Outline a clear, testable research question. [2]

Answer:

How does the temperature of the surroundings (°C) affect the time taken (minutes) for an equal-sized ice cube to melt completely?

IV: Temperature of surroundings

DV: Time taken for the ice cube to melt

Q6.2 Outline a testable prediction and support it using scientific reasoning about particles. [2]

Hypothesis:

If the temperature of the surroundings increases, then the time taken for an equal-sized ice cube to melt completely will decrease.

Scientific reasoning:

At a higher temperature, more **thermal energy** is transferred from the surroundings to the ice. The particles gain **kinetic energy** and vibrate more rapidly. As the ice absorbs enough energy, the particles can overcome some of the **forces of attraction** holding them in fixed positions, causing the ice to **melt faster**.

Higher temperature → faster melting → less time taken

Lower temperature → slower melting → more time taken

Scientific terminology: thermal energy • kinetic energy • particle movement • melting

Q6.3 Outline the variables. [3]

i. Independent variable:

Temperature of the surroundings (°C)

ii. Dependent variable:

Time taken for the ice cube to melt completely (min/s)

iii. Two controlled variables:

- **Size/mass of each ice cube**
- **Type/size of container**

Other acceptable controlled variables include the starting temperature of the ice cubes and the type of water used to make the ice.

Q6.4 Design a logical and safe investigation. [7]

Answer:

Materials/Apparatus:

- 3 equal-sized ice cubes
- 3 identical containers
- Thermometer
- Stopwatch
- Controlled-temperature environments/water baths
- Tongs
- Paper towels

Method:

1. Prepare **equal-sized ice cubes**.
2. Prepare three different temperatures, for example **15°C, 25°C and 35°C**.
3. Measure each temperature using a **thermometer**.
4. Place one ice cube into an identical container at each temperature.
5. Start the **stopwatch** immediately.

6. Record the **time taken for each ice cube to melt completely**.
7. Keep the size of the ice cubes and type of container **constant**.
8. Repeat the investigation and calculate the **mean melting time** for each temperature.
9. Compare temperature with mean melting time.

Safety:

Wipe up melted water immediately to prevent **slipping**. Handle water baths/containers carefully to avoid **burns**.

Criterion C – Processing and Evaluating

Q7. Investigating Dissolving

A student investigates how water temperature affects the time taken for **10 g of sugar to dissolve in 100 mL of water**.

Temperature	Trial 1	Trial 2	Trial 3
20°C	124 s	120 s	122 s
30°C	101 s	99 s	100 s
40°C	78 s	81 s	78 s
50°C	61 s	59 s	60 s
60°C	43 s	41 s	42 s

Hypothesis: If the temperature of the water increases, the time taken for sugar to dissolve will decrease.

Q7.1 Calculate the mean time for each temperature and present the data on an appropriate graph.

Answer:

Temperature	Mean dissolving time
20°C	122 s
30°C	100 s
40°C	79 s
50°C	60 s
60°C	42 s

Graph:

Title: Effect of Water Temperature on Sugar Dissolving Time

X-axis: Temperature of water (°C)

Y-axis: Mean dissolving time (s)

Plot: **(20,122), (30,100), (40,79), (50,60), (60,42).**

Q7.2 Interpret the data and graph to describe the relationship between temperature and dissolving time. Use at least two numerical values and particle theory.

Answer:

As **temperature increases, the time taken for sugar to dissolve decreases.**

At **20°C**, the mean dissolving time is **122 s**, while at **40°C** it decreases to **79 s**. At **60°C**, it decreases further to **42 s**.

At higher temperatures, water particles have greater **kinetic energy** and move faster. This increases interactions between the water particles and sugar, so the sugar **dissolves faster**.

Therefore:

Higher temperature → greater particle movement → shorter dissolving time.

Q7.3 Discuss whether the hypothesis is supported. Use two pieces of numerical evidence and give a conclusion about the validity of the prediction.

Answer:

The hypothesis is **supported by the results**.

At **20°C**, sugar took an average of **122 s** to dissolve, whereas at **40°C**, it took **79 s**. At **60°C**, the time decreased further to **42 s**.

The data shows a consistent pattern: as the **temperature increased**, the **dissolving time decreased**.

Therefore, the prediction is **valid for the conditions tested** because the experimental evidence supports the hypothesis.

Q7.4 Discuss one strength and one limitation of the method.

Answer:

Strength:

Three trials were carried out at each temperature. **Repeated trials** make the results more **reliable**, and calculating a **mean** reduces the effect of random variation.

Limitation:

The information does not state whether the sugar was **stirred at the same rate** in every trial. Different stirring rates could affect the dissolving time and reduce the **validity** of the results.

Improvement: Keep the stirring rate constant in every trial.

Criterion D – Reflecting on the Impacts of Science

Q8.1 Summarize how the science of evaporation is used in salt production. Include at least three relevant scientific points. [4]

In salt production, **seawater is placed in large, shallow ponds called salt pans**. Heat from the **Sun** provides energy to the water. During **evaporation**, some water particles

at the surface gain enough energy to escape from the liquid into the air as **water vapour**.

The salt pans are shallow and have a **large surface area**, which allows more water particles to be exposed to the air and therefore increases the **rate of evaporation**. As the water evaporates, the **dissolved salt remains behind** because the salt does not evaporate with the water. The remaining salt becomes concentrated and can eventually be **collected for use**.

Key scientific terminology: evaporation • water vapour • surface area • solar energy • dissolved salt • rate of evaporation

Q8.2 Describe and summarize the positive and negative implications of using evaporation for salt production in relation to environmental and social/economic factors. [6]

Answer:

There are both **positive and negative environmental implications** of using evaporation for salt production.

One environmental benefit is that the process can use **solar energy** from the Sun to evaporate seawater. Solar energy is a **renewable energy source**, and the evaporation process does not require the burning of **fossil fuels**. This can reduce the environmental effects associated with fuel use.

However, there are also environmental disadvantages. Salt production requires **large, shallow salt pans**, which need a considerable amount of **land**. Creating these salt pans may affect or disturb nearby **habitats**. Therefore, although using sunlight has environmental benefits, the amount of land required can have a negative environmental impact. MYP 1 Integrated Sciences – Ter...

There are also important **social and economic implications**. Salt production can provide **employment** for people and produces salt that is needed for **food and industry**. This makes salt production economically useful to communities.

However, workers may have to spend **long hours outdoors in hot conditions**, which is a negative social implication. Salt production is also strongly dependent on **weather conditions**. During rainy or **humid conditions**, evaporation becomes slower, which reduces the amount of salt that can be produced. Reduced production may affect the economic benefits of the process.

Overall, evaporation is useful for salt production because it uses a **renewable source of energy**, provides **employment**, and supplies an important material for food and industry. However, its benefits must be considered alongside limitations such as **land use, effects on habitats, difficult working conditions and dependence on suitable weather**

For Criterion D, Please follow:

Factor → Implication → Explanation → Evidence/example

For example:

Environmental factor: Solar evaporation uses **renewable solar energy**.

This is beneficial because it does not require burning fossil fuels for evaporation. **However**, large salt pans require considerable land and may affect nearby **habitats**.

That is much stronger than simply writing *"Solar energy is good for the environment."*

Environmental – Positive:

Solar evaporation uses **sunlight**, which is a **renewable source of energy**. It does not require burning fossil fuels for the evaporation process.

Environmental – Negative:

Large **salt pans require considerable land** and may affect nearby **habitats**.

Social/Economic – Positive:

Salt production provides **jobs** and produces salt needed for **food and industry**.

Social/Economic – Negative:

Workers may spend long hours outdoors in **hot conditions**. Salt production is also **weather dependent**. During rainy or humid conditions, evaporation becomes slower, which can reduce salt production.

Overall:

Evaporation is useful for salt production because it uses **renewable solar energy** and provides economic benefits, but it also has limitations related to **land use, habitats, working conditions and weather dependence**.